2.3 Solving Inequalities Using Multiplication or Division

Name _____ Period ___ Date _____

Algebra I 2.3 · Student sheet

I can…

- solve inequalities by multiplying or dividing by a positive number.
- solve inequalities by multiplying or dividing by a negative number.
- write and solve inequalities for real-life problems.

Vocabulary · fill in the missing word

Multiplying or dividing by a positive number — the _____ symbol stays the same

Multiplying or dividing by a negative number — the _____ symbol reverses

Part 1 · Does it flip?

Start with the true statement **4 < 6**. Do the same thing to both sides.

Do to both sides	New statement	Still true?	If not, fix the symbol
× 2			
× (−2)			
÷ 2			
× (−1)			

Rule: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Positive numbers: Multiply or divide by a positive number: _____ the symbol.

Example 1

Solve $\frac{x}{4} < 3$.

Negative numbers: Multiply or divide by a negative number: _____ the symbol.

Example: $-2x \geq 10 \rightarrow x$ ____ -5

Example 2

Solve $-5y \geq 35$.

Example 3

Solve $-\frac{x}{3} > 2$ and graph.

Example 4

Movie tickets cost $7.50 each. You have $60. How many tickets can you buy?

Part 3 · Partner practice

Partner 1

Solve $6n > -24$.

Partner 2

Solve $-0.5m \leq 4$ and graph.